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Chatbot Persona Selection Methodology for Emotional Support

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Abstract: Self-help groups are one way to cope with mental illness. Here, people in similar circumstances come together to provide peer support and help each other. However, some people are not comfortable using such groups. Therefore, it is necessary to investigate the possibility of using robots or other means of support. There are two types of identity: social identity and personal identity. Personal identity is defined as a kind of personality. Social identity is like a persona, a profile, such as a job or position. So far, we have suggested implementing personalities as chatbot identities and adapting the communication method to the user. However, using personalities alone would result in bland content. Therefore, we implement social identity in chatbots and use the chatbot's personal information to enable the chatbot to speak in a way that is sympathetic to the user.

Keywords: Stress management, Interactive GA, Emotional support, Social identity

1. INTRODUCTION

Occupational stress is a significant workplace problem with high costs for individuals and society [1]. A range of factors contribute to an increase in reported stress by workers and include inadequate support from management, low levels of control, inappropriate task allocation [2].

According to a survey undertaken by The Japan Institute for Labour Policy and Training [3], people working in the healthcare sector are at greater risk of stress-related mental health problems than all other professions. The outcomes of exposure to workplaces stressors impacts individuals differently, but can result in depression, health related illnesses, underperformance and increase the potential for early retirement.

Although personal counselling is often used in the treatment of depression by human, support sessions with chatbots are increasing [4]. Sessions with chatbots are aimed at managing emotion by performing cognitive behavioural therapy (CBT) or emotional support. Smith et al. investigated the virtual agent's emotional support and showed that it is not as effective as human support, but it has sufficient functionality as support [5].

With the advent of ChatGPT, ordinary people can now have normal conversations with non-human entities. With chatbots, users can talk about things that are difficult to talk about with humans. Also, in the case of humans, compatibility with a counselor is a factor, but with chatbots, the user can choose the type he or she likes. For example, the SELF application has a variety of characters with different personalities [6]. In general, chatbots choose personalities that users are likely to like, while Alexa is designed to be smart, approachable, humble, enthusiastic, and friendly [7]. XiaoIce's persona is always described as a trustworthy, 18-year-old girl with a great sense of humor [8].

Turner divided self-concept into social and personal identities [9]. Social identity is described by group membership in a social category. Personal identity is an individual's sense of competence, psychological dispositions, abilities, and preferences. What concerns personality, "I am sociable" is a personal identity, while "I am a university student in Japan" is a social identity. In this study, personality corresponds to personal identity, and persona is considered as social identity and implemented as a self-concept of the chatbot. In addition, the chatbot selects a chatbot's self-concept that is close to the user's self-concept by flexibly changing the self-concept during the interaction with the user. The goal is to improve usability in the emotional support of peer support.

Section 2 describes persona interaction systems, interactive genetic algorithms (IGAs), and related research on using chatbots and robots in mental health care. Section 3 provides a brief summary of the stress management framework. Section 4 shows how the chatbot decides the self-concept and interacts with the user. Section 5 presents the experimental results of the proposed method. Section 6 provides conclusions and future work.

2. RELATED WORK

A. Personality in human-robot interaction

In human-robot interaction, personality is considered to be an essential factor in enhancing the effect of interaction, and various studies have been conducted [10]. Personality is considered a set of characteristics that are stable across situations and time and provide guiding influences on an agent's behavior and

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interactions [11]. A persona is also a fictional character, with a name, age, education, occupation, and backstory.

Masuyama et al. used personality and emotion models to make robots generate different outputs for the same input [12]. At this time, personality is generated based on gender and age using the Big Five model. The big Five model consists of Openness to experience, Conscientiousness, Extroversion, Agreeableness, and Neuroticism, and it is believed that a person's personality can be classified according to five factors. It was shown that it is possible to have different interactions by generating different outputs depending on different personalities.

Tapus et al. focused on the extroversion-introversion of a socially assistive therapist robot, and the robot interacts with users with the same personalities [14]. They used Eysenck's personality model, which has three dimensions: Extraversion, Neuroticism, and psychoticism, and behavioral adaptation was performed using a reinforcement learning algorithm.

Reeves and Nass describe that agents have the same personalities as people to enhance the effect of interaction, and it is more effective to gradually have the same personalities as people even though they initially have different personalities [13]. In interaction, it is essential to match personalities, as it is highly appreciated when communicating with a person with the same personality. It is considered inflexible if personality is fixed at the first stage.

Conversely, in another experiment, which examined whether humans like robots to have similar or different personalities to themselves, it was shown that humans seek robots with different personalities from their own [15]. It has been proposed that the task decides whether a human will seek similar or complementary personality traits, but this has not been proved [16]. The choice may also depend upon whether participants have stereotyped ideas about robots and how they should be.

B. Persona-based systems

Persona dialogue systems provide chatbots with profile information in the form of personas to ensure consistent dialogue in line with their personalities. In persona chat, personas are fixed and cannot be automatically updated based on response history [17]. Systems that add new personas to existing personas have been studied to address this problem.

Robots with multiple personas include service robots and social robots. In the case of robots, they do not engage in conversation, but rather express their personas by performing actions that match their personas [18]. The service robot example shows that multiple personas improve exploratory performance in new environments. multiple personas in HRI were proposed to design complex and nuanced interactions in robots taking on roles such as receptionist or nurse [19]. Olly used the Big Five characteristics created robot personalities and personas, suggesting that extreme and positive characteristics would be ideal [20].

3. STRESS MANAGEMENT FRAMEWORK

We are developing a stress management framework to support people at work in managing their stress levels [21]. It uses different agents depending on the location so that the user can have the agent as a peer at any time in which they require support. As shown in Figure 1, robots are used at home and the workplace, and a chatbot is used for all other places between home and work.

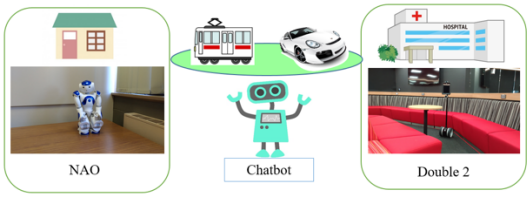

Figure 1. System configuration

The agent has software architecture, as shown in Figure 2 for stress management. The user's stress condition is recognized by evaluating the input content from the user based on Sense of Coherence (SOC), and the appropriate support is selected and the interaction follows the conversation model.

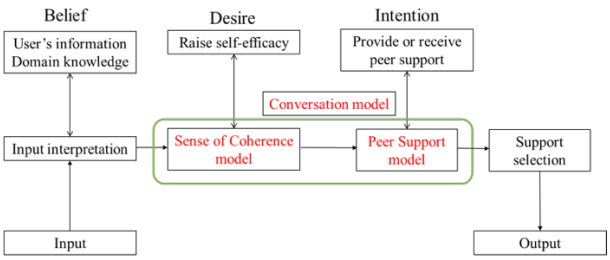

Figure 2. BDI architecture within our stress management framework

The conversation model consists of five phases, as outlined in figure 3. Phases 1 to 3 were the subject of a previous paper [21]. In this paper, we implement phase 4 and 5 and describe initial results.

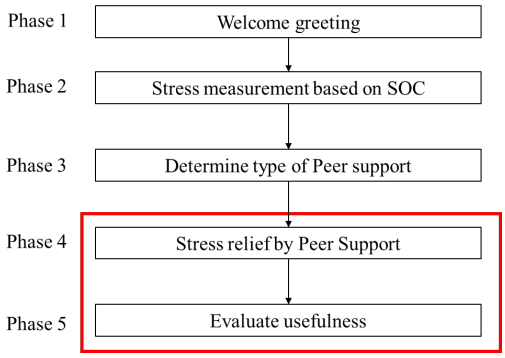

Figure 3. 5-phase conversation model

The support which the agent offers in phase 4 depends on the value of the SOC. If the SOC is low the agent imitates a peer support method. However, if the agent evaluates the users SOC as high, then agent asks the user for support. Each peer support method corresponds to a certain combination of SOC parameters, Comprehensibility (Co), Manageability (Ma) and Meaningfulness (Me) [21]. Emotional support

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is selected if the agent evaluates that Meaningfulness is low (Fig.4).

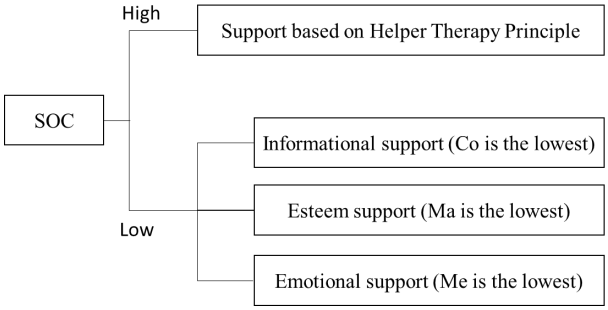

Figure 4. How to decide the support

We used a chatbot for smartphones, considering the ease of conducting experiments because many chatbots are developed for smartphones. To develop the chatbots, we used the LINE chatbot platform. LINE is a free social networking service (SNS) application that provides an API for creating chatbots. the LINE Messaging API is an application programming interface that enables user interaction and conveys user input to the LINE application server [22]. The system configuration is shown in Figure 5.

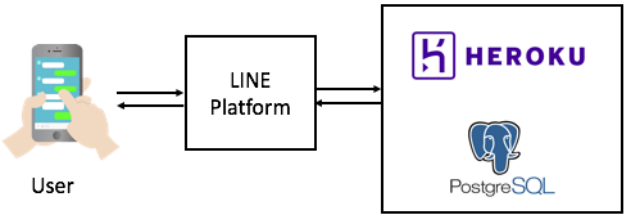

Figure 5. Chatbot system configuration

The user communicates with the chatbot by entering free text or by selecting a fixed response option from a menu presented by the chatbot. The chatbot then processes the user's input and responds according to a set of logic. Information about the user's stress state and the chatbot's conversational state is stored in PostgreSQL.

4. EMOTIONAL SUPPORT USING PERSONA

A. Self-Categorization of Chatbots

Because persona and personality are sometimes used in the same sense, this study distinguishes between the two using the ideas of self-categorization theory [9]. Self-categorization considers three levels of how social self-concept is categorized in relation to others (Fig.6). We use two levels: social identity as the upper level of being human, and personal identity as the lower level. In this study, personal identity is used as personality and social identity as persona.

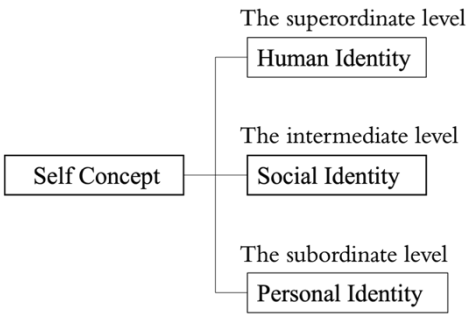

Figure 6. Three levels of self-concept

B. Interactive Genetic Algorithm for user adaptation

One example of Interactive Evolutionary Computation (IEC) is the Interactive Genetic Algorithm (IGA). We proposed IGA to adapt the chatbot's personality to reflect the user's personality preference. IGA replaces the traditional fitness function of a Genetic Algorithm with human subjective judgment. This enables an IGA to accommodate user preferences that are difficult to quantify, such as likes, dislikes and sensitivity, there are many examples of studies embedded in design systems [23]. A potential drawback of IGA is their time to convergence since the fitness function requires user input. However, the system under optimization can adapt while in use.

Our chatbot personality model uses two key personality traits from the Big Five model: Extroversion and Agreeableness. We focus on the two traits because they are necessary for the chatbot's interaction. Extroversion is the tendency to be energetic, sociable, talkative and to seek stimulation of others. Agreeableness tends to be a personality trait which is caring and supportive of others. High agreeableness is often seen as naive or obedient. In our chatbot personality model, extroversion is used to determine the number of conversations with the user. Agreeableness determines whether the chatbot's behaviour is dominant or submissive. The gene pool population size is G (=10), and the chromosome length is L (=2), corresponding to the number of personalities (Extroversion and Agreeableness). The ith individual is represented by ($g_{i,1}$, $g_{i,2}$). The gene uses a real type and takes a numerical value between 0 and 1, respectively, where 0 represents low Extraversion or low Agreeableness, and 1 represents high Extraversion or low Agreeableness.

Our proposed method is shown (Fig.7). The initial gene pool of personalities is randomly generated, and one gene is randomly selected by the current personality for the chatbot. The user then talks with the chatbot from phase 1 to phase 5. At the end of phase 5, the user assesses the interaction, and a genetic operation is performed based on the user's feedback.

The user evaluates the interaction with a value of 1 to 10. If the evaluation is 10, the current personality gene becomes the elite gene in the next generation. The elite gene then takes over the personality of the chatbot. In addition, mutation operations are performed on individuals other than the elite. The mutation here is a

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uniform mutation, made by adding or subtracting a random number from 0 to 0.5.

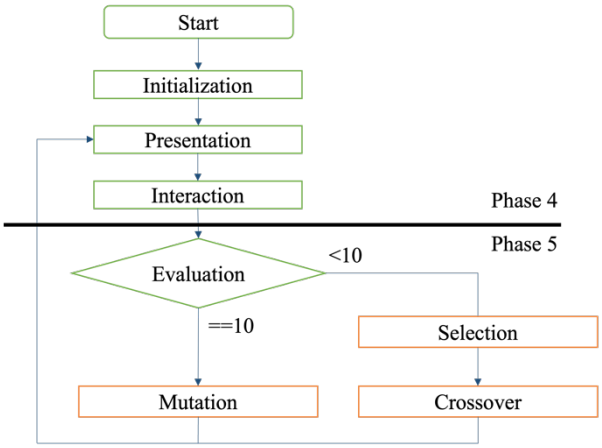

Figure 7. Flowchart of Interactive genetic algorithm in the framework

If the evaluation is less than 10, the selection is made depending on the evaluation value. The fitness value of each individual is expressed as the distance of personality from the current personality gene.

$$fit_i = \sqrt{\left(g_{display,1} - g_{i,1}\right)^2 + \left(g_{display,2} - g_{i,2}\right)^2} \quad (1)$$

Next, we determine the next generation's elite, as shown in figure 8.

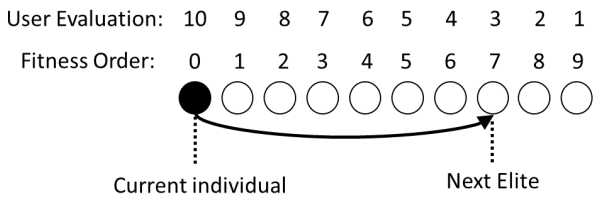

Figure 8. The Method of Elite Selection

For example, if the evaluation from the user is 3, the next elite is the seventh individual when arranging the individuals in order of fitness values (Fig.8).

Then crossover will be performed between the elite and those chosen randomly. Blended Crossover (BLX-a) is used as the crossover method [24]. The generated child replaces the displayed individual. In BLX-a, the element of the child is decided randomly between x and y.

$$x = \min\left(g_{elite,j}, g_{i,j}\right) - \alpha d_j \quad (2)$$

$$y = \max\left(g_{elite,j}, g_{i,j}\right) + \alpha d_j \quad (3)$$

$$d_j = \left|g_{elite,j} - g_{i,j}\right| \quad (4)$$

Personas are determined from the selected personalities. The method of persona selection will be based on Jung's typology. In Jung's theory, there are eight types consisting of two categories, extroversion and introversion, and four functions: thinking, feeling, sensing, and intuition. In this study, extroversion and cooperativity will be used to classify the personas into thinking and feeling personas. The thinking function is considered less cooperative, and the emotional function is considered more empathic.

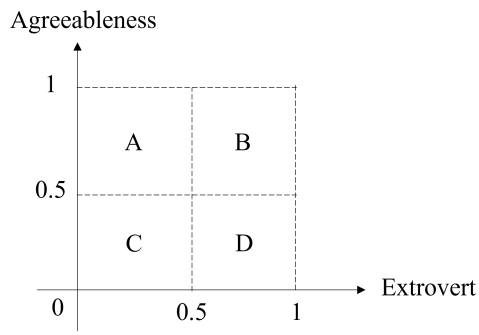

Figure 9. Selection of personas

Now, based on the personality values, we can apply personas A-D. The x-axis is extroversion and the y-axis is cooperativeness; when x is large, the person is extroverted and when y is small, the person is introverted. Also, when y is large, the person is an emotional type, and when y is small, the person is a thinking type; A is an introverted emotionalist, and D is an extroverted thinker. According to the literature [27,28], A is Chopin, B is Frank Sinatra, C is Kant, and D is Darwin.

C. Emotional Support

In this system, the chatbot engages in actively listening to the user's story using repetition, confirmation, sympathy, and support messages. It uses Sentiwordnet [25] as emotion recognition, giving persona-specific messages when the analysis is positive, and repetition, confirmation, and sympathy when it is negative. Repetition involves repeating the user's input. Confirmation uses questions to ensure that the user's input is understood. Sympathy provides a comforting response. The categories of Emotional Support are as described by Smith et al [26]. Their flowchart is shown in Figure 10.

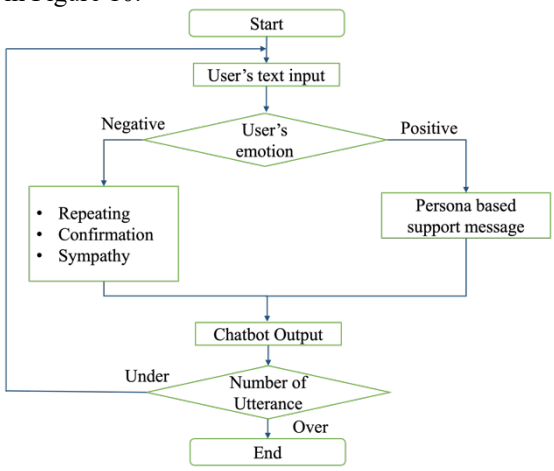

Figure 10. Flowchart of the chatbot's user interaction

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5. EXPERIMENTAL RESULT

We demonstrate that assistance can be provided by interacting with the chatbot, and changing personas. The initial persona is selected at random. Here, interactions are conducted with the goal of producing a persona opposite to the initial persona, and after several interactions, we investigate whether the goal can be achieved. Conversations are implemented by adding persona features to active listening in emotional support when no persona is present. In the experiment, we tested whether the persona could be changed according to the user's evaluation.

TABLE I. EXPERIMENT CONDITION

Name	Content
Input	• User conversation • User evaluation of conversation from 1 to 10
Output	Reply text to user based on probability
Internal state	Random values

A. Experiment scenario

We assume that phase 2 of the conversation model has been evaluated, and the users' SOC indicates low meaningfulness (Me). As a result of this, the chatbot proposes emotional support in phase 3, and the user receives support in phase 4.

The user will converse with the chatbot for a period controlled by the chatbot's personality, termed conversation episodes. After each episode, the chatbot will ask the user to rate the conversation. This rating is used as a fitness function in the IGA. We expect initial user evaluations will be low to medium. With regard to the convergence, IGA terminates when the user is satisfied, but in this case, it is judged when an individual having the same personality emerges continuously. We use a command prompt on a personal computer to talk with the chatbot program.

B. Experiment results

Figure 9 shows the number of elite individuals. Since we were trying to see if we could use all the personas, all the evaluation values were given 1. When 1 was given, the individual with the farthest property from the current elite was chosen as the next elite, so the elites changed each time. The first persona that appeared was Chopin, and then Darwin and Frank Sinatra appeared alternately. Kant never appeared. This is because the initial values of the individuals are randomly determined, which may make it impossible for them to appear in some situations.

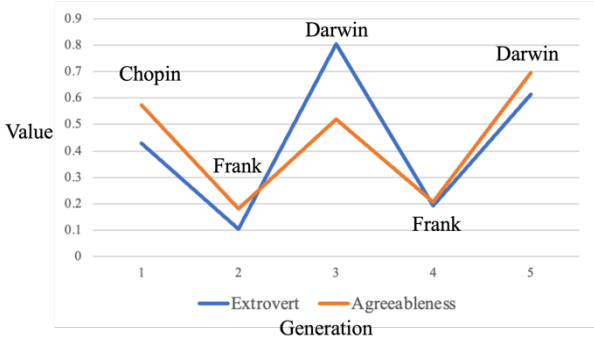

Figure 11. Persona history

An example of the dialog is shown in Figure 12. Chopin is selected as the persona. The selected content is random, so the user must adapt to the content of the conversation, but the chatbot has determined that persona messages may be used successfully; messages with PER at the end of the sentence are persona messages, but they need to be adjusted so that they are not used too often. Persona dialogue has allowed us to interact with information that would not have come up in dialogue without personas. Without personas, the dialogue would be bland, but with personas, it is possible to incorporate internal elements of the individual into the dialogue and create empathy with the user.

It is also possible to switch between personas while conducting a dialogue. Whether all personas can be used depends on the initial distribution of individuals and needs to be refined.

Chatbot	>> Please talk about your anxiety.
User	i can't sleep well last night
Chatbot	>> Oh, how miserable it is to have no one to share your sorrows and joys, to not be able to share your true feelings when your heart is heavy.(PER)
User	I was very sad
Chatbot	>> Wow that must have been hard.
User	I want to be happy
Chatbot	>> I create a new world for myself.(PER)
User	so great
Chatbot	>> Music is only what your ears allow.(PER)
User	I think so
Chatbot	>> Thanks for sharing. How do you rate this support?[1:10]

Figure 12. Conversation example

6. CONCLUSION

We developed and experimented with a system for persona matching for emotional support. For personas, we used the characters described in Jung's type theory as they are. Still, in the future, we will create unique personas for counseling that have the characteristics of these characters so that we can have conversations that match the purpose. The conversation method is random, so we need to consider a system using Large Language Model (LLM), such as ChatGPT.

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